



Smart Retail Analytics & Demand Forecasting using Machine Learning

B.Tech Information Technology — Submitted by: Pramit Shrivastav

Problem Statement

Retailers collect vast daily sales data that remains underutilised, causing inaccurate demand prediction, poor inventory planning, overstocking, shortages and inefficient decisions.

Objectives

Analyse historical sales, identify trends & customer behaviour, forecast demand with ML, improve inventory, deliver an interactive web dashboard.



Solution Overview

An end-to-end retail analytics platform built with Python, Flask and Machine Learning to transform sales data into actionable forecasts and insights.



Dataset Upload

Retail dataset upload with automated ingestion.



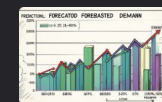
Preprocessing

Automated cleaning and feature engineering.



Analytics & Visuals

Sales/customer analytics and interactive visualisations.



Demand Forecasting

Machine learning models produce future demand estimates for planning.

Project Workflow



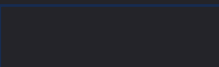
Data Prep

Feature Build

Explore

Forecast & Dash

Each stage transforms raw sales data into forecasts and dashboard insights for decision-makers.



Technology Stack



Programming

Python



Backend

Flask



Frontend

HTML, CSS, Bootstrap, JavaScript



ML Tools

Scikit-learn, XGBoost, Pandas, NumPy



Visualisation

Matplotlib, Seaborn



Database

SQLite





Machine Learning Pipeline

01

Data Collection & Cleaning

Gather and prepare sales data for modelling.

02

Feature Selection & Training

Engineer features and train models (e.g., XGBoost).

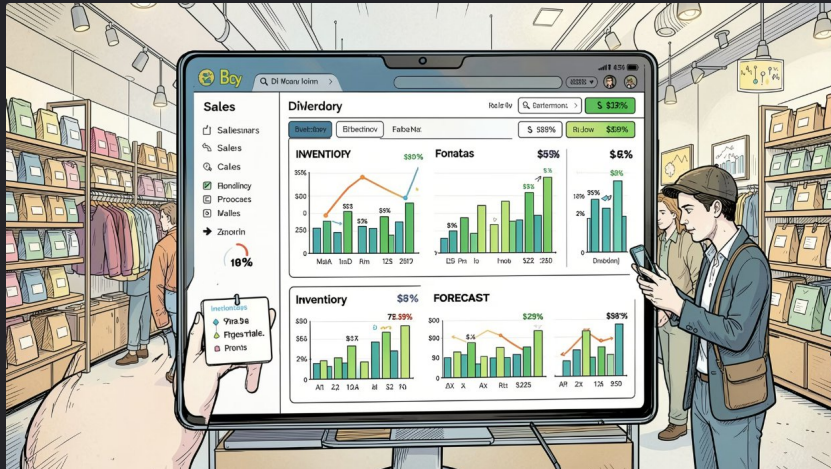
03

Evaluation & Prediction

Evaluate models and generate demand predictions.

Evaluation Metrics: MAE, MSE, RMSE, R^2 Score

Features & Dashboard



Sales Analytics

Detailed sales trends and product performance.



Customer Analytics

Behavioural insights to segment and target customers.



Inventory Insights

Stock optimisation guided by forecasts.



Reports & Exports

Interactive charts and business reports for stakeholders.

Results & Benefits

Results

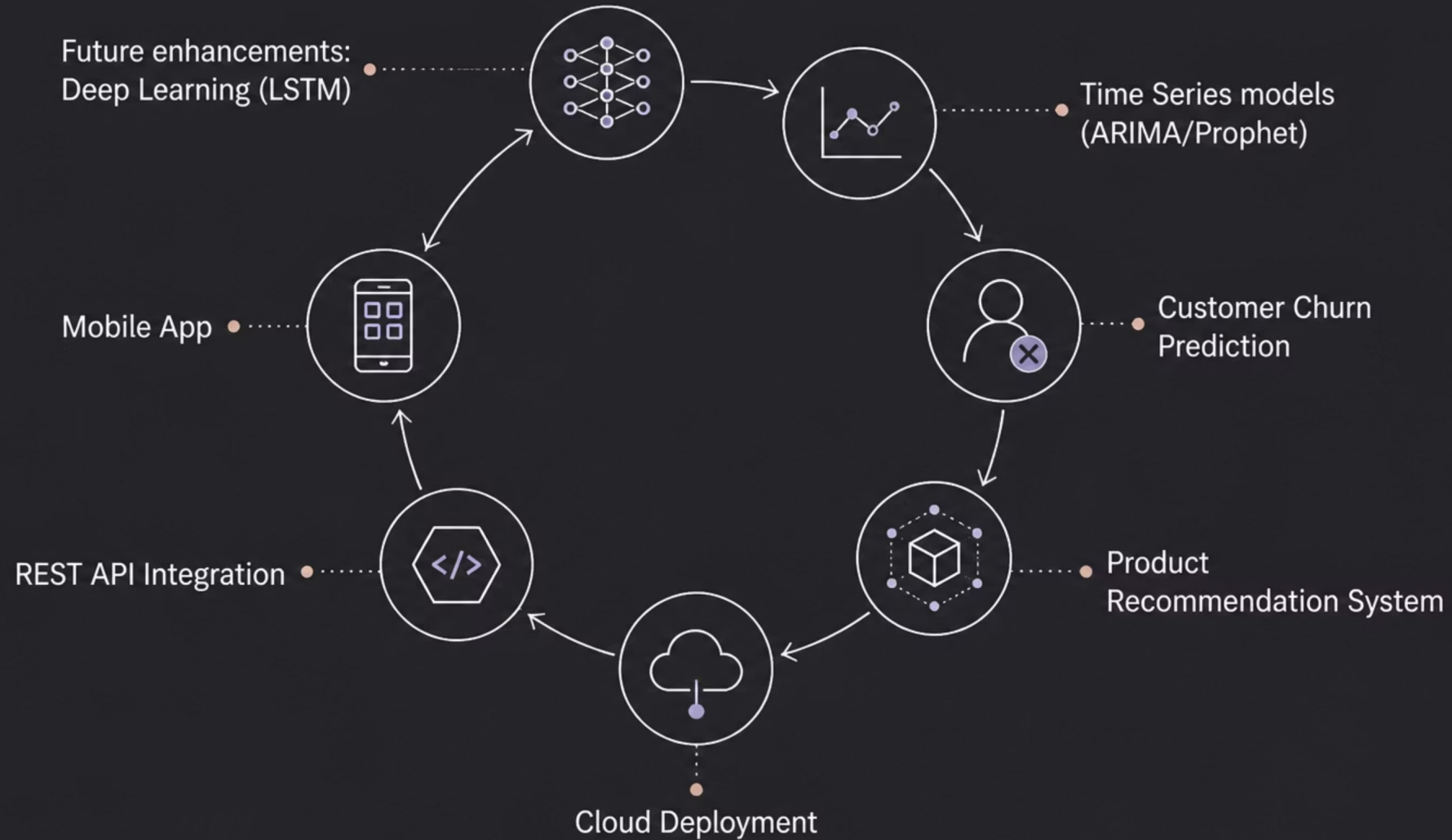
System analyses sales, identifies trends, predicts demand, and generates interactive visualisations to support inventory planning.

Benefits

Improved decision-making, reduced manual analysis, better inventory optimisation and more accurate sales forecasting.



Future Scope



Next Steps

Prototype to Production

Deploy models to cloud with REST APIs and integrate real-time data feeds.

Advanced Modelling

Introduce LSTM and Prophet for improved time-series forecasting.

Productisation

Build mobile app, recommendation modules and customer churn features.

- ❑ Focus on scalable deployment and continuous model evaluation to maximise business impact.

